

Oscillator and PLL

This chapter describes the oscillator and PLL clock source paths for the device.

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10.1 Introduction

This chapter provides an overview of the oscillator and PLL clock source paths for the device.

The oscillator macro will pass a signal driven into the OSCIN pin to clock source 0 that is the device default clock source on reset. When a crystal or resonator with appropriate load circuitry is connected to OSCIN and OSCOUT, the oscillator macro drives the crystal/resonator to generate the input waveform. In addition to being directly usable as clock source 0, the oscillator clock is the input to the PLL.

The oscillator frequency is continuously monitored by a dedicated clock detect circuit. If the frequency falls out of a fixed range, the clock detect switches the clock from the oscillator to an internally generated, free-running frequency (generated by the low-power oscillator (LPO)).

The phase lock loop (PLL), a circuit in the microcontroller, is used to multiply the input frequency to some higher (device operation) frequency. This frequency synthesis is useful for generating higher frequencies than can be conveniently achieved with an external crystal or resonator. Additionally, the PLL allows the flexibility to be able to synthesize one of multiple frequency options from a given crystal or resonator.

Frequency modulation can be superimposed on the synthesized frequency. The modulation provides a means to reduce the impact of electromagnetic radiation from the device; this reduction in measured radiation can be useful in sensitive applications.

10.1.1 Features

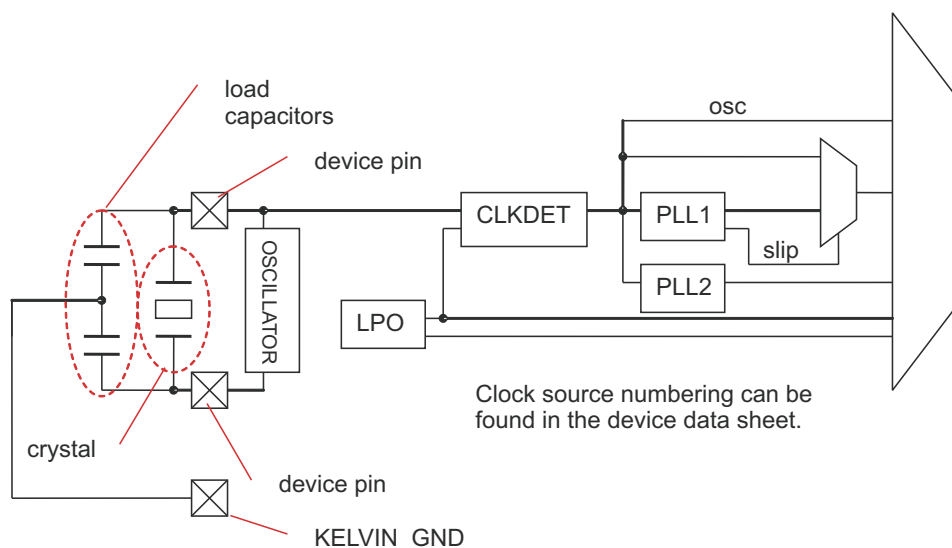
The main features of the source clock path are:

- The oscillator may drive a crystal/resonator or be driven from an external source
- The clock detect provides continuous monitoring of the oscillator frequency and provides an automatic switch over to a free-running clock in case of oscillator failure.
- The FM-PLL module can be operated in either modulation or non-modulation mode.
- The phase-frequency detector assures lock to the fundamental reference frequency.
- $$f_{PLL} = \frac{f_{OSCIN}}{NR} \times \frac{NF}{OD \times R}$$
 (1)
 - Configurable prescale divider (NR) for the input clock
 - Configurable multiplier (NF)
 - Configurable postscale dividers (OD, R)
- The PLL may be used with modulation enabled.
 - Configurable modulation frequency (NS)
 - Configurable modulation depth (NV)
- The slip control circuitry provides flexible response to a PLL failure (slip) including reset or automatic switch over to oscillator.

10.2 Quick Start

The purpose of this section is to provide an overview of how to configure the oscillator and PLL clock paths on power-up. More detailed descriptions are presented in later sections. [Figure 10-1](#) shows the oscillator and PLL clock paths.

Figure 10-1. Clock Path From Oscillator Through PLL To Device



While power-on reset is asserted (low), the oscillator and low-power oscillator (LPO) are enabled and start-up by default. After power-on reset is released to a high level, the clock detect circuit (CLKDET) begins to monitor the oscillator. If the oscillator is within a valid range, the oscillator becomes the default clock for the device as it exits reset; if the oscillator is not within a valid range, the clock detect selects the high-frequency low-power oscillator as the default clock for the device.

The low-power oscillator has a wide frequency range which also creates a large valid window for the clock detect; in order to refine the clock detect window, the low-power oscillator can be trimmed. The initial trim value is stored in one-time programmable section of the flash memory, address 0xF008_01B4. Bits 31:16 of this word contain a 16 bit value that may be programmed into LPOMONCTL(15:0) in order to initialize the trim for both HF LPO and LF LPO. Software should read the initial trim values from flash and write them to the control register.

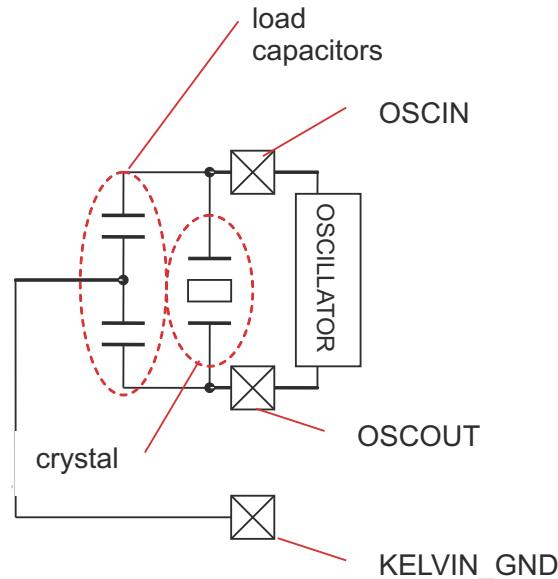
The PLL is disabled by default on power-up. The PLL control registers (PLLCTL1 and PLLCTL2) must be configured to set the desired output frequency. Then, the system PLL may be enabled (CSDISCLR.1). Similarly, the second PLL must be configured in PLLCTL3 and enabled (CSDISCLR.6). Each PLL has a valid bit that indicates the PLL is locked (CLKSRnV bit in the Clock Source Valid Status Register (CSVSTAT) of the System and Peripheral Control Registers).

Prior to selecting the PLL clock as the source for a clock domain (for example, GCLK, HCLK, VCLKA1), the domain and modules on the domain must be configured to accept the new frequency. An example of a module that should be configured prior to selecting the PLL as clock source for GCLK and HCLK is the memory wrapper to insure that access times are maintained correctly.

10.3 Oscillator

The clock generation path through the PLL begins with the oscillator. The oscillator consists of three separate pads -- OSCIN, OSCOUT, and Kelvin_GND (see [Figure 10-2](#)).

Figure 10-2. Clock Generation Path



The oscillator is responsible for two independent functions:

1. The oscillator is responsible for generating positive feedback in the external crystal/resonator with appropriate load and tank circuitry. At start-up, the oscillator amplifies random noise. The external circuitry acts like a band-pass and selects the crystal/resonator frequency to provide as positive feedback into the amplifier. The positive feedback increases the amplitude of the output waveform into the crystal/resonator (and the load circuitry), and the voltage waveform shows an envelope of increasing amplitude. The oscillator can drive a crystal frequency that is within the data sheet range $t_{c(OSC)}$.

Looking at the input waveform into OSCIN, the voltage waveform is an AC-coupled, filtered version of the OSCOUT waveform. The band-pass functionality of the crystal/resonator removes distortion from the OSCOUT waveform, leaving a sinusoidal input waveform.

NOTE: Vendor Validation of Resonators/Crystals

The crystal is a very tight bandpass filter while a resonator is a somewhat wider bandpass. The load circuitry pulls the center frequency of the bandpass.

Texas Instruments strongly encourages each customer to submit samples of the device to the resonator/crystal vendor for validation. The vendor is equipped to determine what load capacitances will best tune their resonator/crystal to the microcontroller device for optimum start-up and operation over temperature and voltage extremes. The vendor also factors in margins for variations in the microcontroller process.

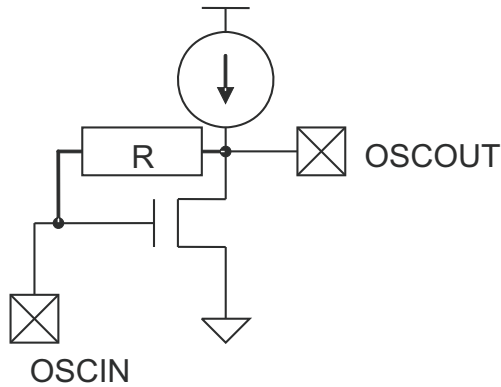
2. The oscillator is also responsible for squaring-up the input waveform. This squaring-up converts the sinusoid into a square wave at the core logic levels. The input path limits the input frequency range as a low-pass filter with a cutoff frequency.

The oscillator has a frequency range that is determined by the driving capability of external crystals/resonators (feedback path). If a clock is driven directly into the oscillator, then the feedback path is not relevant and the frequency range is determined solely by the forward path (which typically allows a higher frequency); the device can support inputs within the data sheet range $t_{c(OSC_Sqr)}$.

10.3.1 Oscillator Implementation

The oscillator operates at 3.3V and uses a constant current source to drive current onto the OSCOUT node. An internal transistor shunts the current (and current from the external circuitry) to GND. This current steering drives the voltage waveform on OSCOUT.

Figure 10-3. Oscillator Implementation



10.3.2 Oscillator Enable

The oscillator is enabled asynchronously when nPORRST is low.

The oscillator is enabled by clearing bit 0 in the Clock Source Disable Register (CSDIS) or setting bit 0 in the Clock Source Disable Clear Register (CSDISCLR) of the System and Peripheral Control Registers. The bit sends a start signal to the oscillator. Bit 0 of CSDIS is cleared to 0 by default on a system or power-on reset so that the oscillator starts-up by default. After the oscillator swings at a high-enough amplitude to pass an input clock into the core domain and nPORRST is released, 1024 oscillator periods are counted before setting the CLKSR0V bit in the Clock Source Valid Status Register (CSVSTAT) of the System and Peripheral Control Registers. The oscillator generates clock source 0 in the global clock module (GCM).

10.3.3 Oscillator Disable

The clock sources (for example, OSC, PLL) are disabled by setting the appropriate bit in the Clock Source Disable Register (CSDIS) or setting the appropriate bit in the Clock Source Disable Set Register (CSDISSET) of the System and Peripheral Control Registers. These bits *allow* the clock source to disable but do not force the behavior until the clock is no longer used as the source for a clock domain (for example, GCLK, VCLK, VCLK2, RTICK). The CLKSR0V bit in the Clock Source Valid Status Register (CSVSTAT), of the System and Peripheral Control Registers, is cleared after clock disable is asserted (which occurs after all clock domains are stopped).

The oscillator disable signal places the oscillator into a low-power state, disconnects the feedback (bias) resistor between OSCIN and OSCOUT, and OSCIN is grounded.

10.4 Low-Power Oscillator and Clock Detect (LPOCLKDET)

The low-power oscillator (LPO) is comprised of two oscillators -- HF LPO and LF LPO -- in a single macro. The low-power oscillator and clock detect (LPOCLKDET) uses a relaxation oscillator to generate an internal clock whose frequency is NOT tightly controlled. This frequency is used to monitor the oscillator input frequency and is also available as an independent clock source in the GCM.

The LPO produces two frequencies:

- High-frequency low-power oscillator (HF LPO) with a nominal frequency of 9.6MHz and a range from 5.5MHz to 19.5MHz; the HF LPO generates clock source 5 in the GCM.
- Low-frequency low-power oscillator (LF LPO) with a nominal frequency of 85kHz; the LF LPO generates clock source 4 in the GCM.

A single current source drives current onto a capacitor; when the voltage on the capacitor exceeds some threshold, the clock toggles. The LPO uses a single current source and the two different comparators to generate the HF LPO and LF LPO frequencies. The LPO is controlled by 4 different bit fields -- CSDIS.(5:4), HFTRIM(4:0), LFTRIM(4:0), and BIASEN.

- CSDIS.5 enables/disables the comparator that generates HF LPO.
- CSDIS.4 enables/disables the comparator that generates LF LPO.
- The HF TRIM and LF TRIM bit fields vary the current into the comparator to independently trim the HF LPO and LF LPO frequencies.
- BIAS ENABLE (LPOMONCTL.24) enables/disables the current source that drives the LPO.

10.4.1 Clock Detect

The LPO HF clock frequency is typically near 9.6MHz but ranges from 5.5MHz to 19.5MHz. The clock detect establishes a window for the oscillator by:

$OSCIN > HF\ LPO_{min} / 4$	$OSCIN / 4 < HF\ LPO_{max}$
$OSCIN > 5.5[MHz] / 4 = 1.375[MHz]$	$OSCIN < 4 \times 19.5 = 78[MHz]$

The clock detect circuit works by checking for a rising edge on one clock (oscillator or HF LPO) between rising edges of the other clock. The result is that in addition to flagging incorrect, repeating frequencies, the circuit also fails due to transient conditions.

The low end of the clock detect window ignores a transient low phase of at least 12 HF LPO cycles.

NOTE: Clock Detection of Oscillator MUST be Disabled Before Disabling HF LPO

The HF LPO frequency is the comparison frequency for the oscillator. The clock detection must be disabled prior to disabling the HF LPO frequency.

If the clock detection is NOT disabled prior to disabling the HF LPO, the clock detect circuitry will fail the oscillator as too fast (compared to the non-existent HF LPO). The clock detect circuitry will switch to the non-existent clock, leaving the device without a valid clock.

10.4.2 Behavior on Oscillator Failure

If the oscillator frequency fails, the clock detects supplies:

- the HF LPO clock to GCM clock source 0 instead of the oscillator
- the HF LPO clock to GCM clock source 1 instead of the PLL

The HF LPO signal will be available as three different clock sources:

- GCM clock source 0 (replacing the oscillator)
- GCM clock source 1 (replacing the PLL)
- GCM clock source 5 as HF LPO

The automatic switch-over from oscillator to HF LPO allows the application to execute at a reduced frequency and respond to a problem with the external crystal/resonator. During and after an oscillator failure, the oscillator CLKSRnV bit in the Clock Source Valid Status Register (CSVSTAT), of the System and Peripheral Control Registers, is set along with the OSCFAIL flag in the Global Status Register (GLBSTAT), of the System and Peripheral Control Registers.

It is useful to explicitly change the GHVSRG register, defining the current clock source for GCLK/HCLK/VCLK domains, to the HF LPO after an oscillator failure.

When reset on oscillator failure is set, PLLCTL1.23 (ROF), the device responds to an oscillator failure by generating a device reset.

10.4.3 Recovery from Oscillator Failure

If the oscillator fails, the clock detect switches the HF LPO frequency onto the oscillator source into the GCM. The OSCFAIL flag in the Global Status Register (GLBSTAT), of the System and Peripheral Control Registers, is also set.

The oscillator may be re-enabled (though if the failure was caused by a hard-fault, the re-enable will fail) through the following procedure:

1. Switch all clock domains from the oscillator to the HF LPO (for example, GHVSRG uses HF LPO, VCLKAn uses HF LPO or VCLK, and so on).
2. If the PLL is used, disable the PLL by setting the appropriate bit in the Clock Source Disable Set Register (CSDISSET) of the System and Peripheral Control Registers.
3. Disable the oscillator by setting the appropriate bit in the Clock Source Disable Set Register (CSDISSET). This action resets the clock detect and allows the oscillator to propagate through GCM clock source 0.
4. Re-enable the oscillator by setting the appropriate bit in the Clock Source Disable Clear Register (CSDISCLR) of the System and Peripheral Control Registers.
5. Clear the OSCFAIL flag in the Global Status Register (GLBSTAT) by writing a 1 to the bit. The PLL slip bits may also be set on an oscillator failure. These can also be cleared.
6. Switch the clock domains back to the oscillator.
7. Re-enable the PLL by setting the appropriate bit in the Clock Source Disable Clear Register (CSDISCLR).

NOTE: Clock Re-Enable Procedure Will Fail If Caused by a Hard Failure

Although it is possible to re-enable the oscillator after a failure, if the oscillator failure was caused by a hard fault (for example, disconnected crystal/resonator terminal), the re-enable process will fail.

10.4.4 LPOCLKDET Enable

The LPO is enabled by default while nPORRST is low. During this time, the current source initializes, holding the relaxation oscillator in reset until initialized. After the current source releases the HF LPO and the LF LPO, these clock frequencies slew to their final frequencies; the final frequency may be achieved while nPORRST is active or after its release. After, nPORRST is released, the HF LPO Valid signal is set 32 HF LPO clock cycles later.

The clock detect is enabled once the oscillator and HF LPO are valid. Because an oscillator failure could occur from reset, the clock detect logic must provide an override path. If the HF LPO is valid and the oscillator is not valid, the clock detect circuitry will become active (overriding the oscillator invalid signal) after 16K LF LPO cycles (about 200ms).

10.4.5 LPOCLKDET Disable

10.4.5.1 Disable Clock Detect

It is possible to disable the clock detect circuitry. For protection, this clock detect disable employs a 2-bit key:

- RANGE DET ENA SSET (CLKTEST.24) must be set to 1
- RANGE DET CTRL (CLKTEST.25) must be cleared to 0

In this case, the LPO HF and LF clocks are still active but the clock detect circuitry is disabled. The clock detect unconditionally switches GCM_CLK_SRC(0) back to the oscillator so care should be taken to insure that the oscillator is good before disabling the clock detect circuitry.

10.4.5.2 Disable LPO HF and LF Clocks

The LPO may be disabled by holding the relaxation oscillator clocks (HF and LF) in reset. The clock detect must be disabled, and any clock domains using either HF or LF clocks must be switched to a different clock source. The LPO HF clock is reset by setting CSDIS.5; CSDISSET.5 is an easy way to set specific bits without disturbing the rest of the register. The HF LPO clock disables several HF LPO cycles after CSDIS is set.

Similarly, the LPO LF clock is reset by setting CSDIS.4, and in a similar way CSDISSET.4 can set the specific CSDIS register bit without using a read-modify-write construction. The LF LPO disables several LF LPO cycles after CSDIS is set.

Restarting the LPO clocks from this condition is fast and is known as a warm re-start. The CSDISCLR register allows the user to clear CSDIS bits without using a read-modify-write code-construct.

10.4.5.3 Disable LPO Current Bias

The LPO current source may be disabled after the clock detect is disabled and HF and LF clock sources are disabled. Turning off this current source places the LPOCLKDET into its lowest power configuration. The bias may be disabled by clearing the BIAS ENABLE bit (LPOMONCTL.24).

Restarting the LPO when the bias current has been disabled requires the current source to initialize first and is, therefore slower than a warm re-start; re-enabling the LPO from this condition is known as a warm re-start (similar to what happens during nPORRST active).

10.4.6 Trimming the HF LPO Oscillator

The HF LPO range varies considerably around 9.6MHz from device to device. In order to provide tighter monitoring of the crystal/resonator, it is useful to trim the oscillator. During device test, a trim value is written into the one-time programmable section of the flash memory (OTP), address 0xF008_01B4. Bits 31:16 of this OTP word contain a 16 bit value that may be programmed into LPOMONCTL(15:0) in order to initialize the trim for both HF LPO and LF LPO.

When trimming the HF LPO, it is recommended to step the trim value so as not to make a large change to any TRIM setting.

After the initial trim, further trimming may be done in LPOMONCTL, using the dual clock compare module (see [Chapter 11](#)) in order to determine the resultant frequency. This module allows for comparison of two clock frequencies. Once the HF LPO is determined to be in-range with the initial HFTRIM setting from the OTP, the crystal oscillator may be used as a reference against which the HF LPO and LF LPO may be further adjusted.

10.5 PLL

The following bit fields from PLLCTL1 and PLLCTL2 configure the PLL:

- REFCLKDIV[5:0]
- PLLMUL[15:0]
- ODPLL[2:0]
- PLLDIV[4:0]
- SPR_AMOUNT[8:0]
- SPREADINGRATE[8:0]
- FMENA

The PLL is responsible for synthesizing an output frequency from the input clock (from the oscillator); [Figure 10-4](#) shows a simple block diagram of the PLL. The FM-PLL divides the reference input for a lower frequency input into the PLL ($f_{\text{INTCLK}} = f_{\text{CLKIN}}/\text{NR}$). The PLL multiplies this internal frequency by NF to get the VCO output clock frequency ($f_{\text{Output CLK}} = f_{\text{INTCLK}} \times \text{NF}$). The PLL output is subsequently divided by two prescale values (OD and R). The value of OD is an integer from 1 to 8 and R is an integer from 1 to 32. This output clock, PLL CLK, sources GCM clock source 1. Valid frequencies are shown in [Table 10-1](#) while [Table 10-2](#) shows how that encoding is generated from the PLL bit fields.

[$f_{\text{(post_ODCLK)}}$ and $f_{\text{(GCLK)}}$ are data sheet parameters.]

Table 10-1. Valid Frequency Ranges for PLL

	Frequency Limit
f_{CLKIN}	$f_{\text{(OSC_Sqr)}}$
f_{INTCLK}	1MHz - $f_{\text{(OSC_Sqr)}}$
$f_{\text{Output CLK}}$	150MHz - 550MHz
$f_{\text{post-ODCLK}}$	$f_{\text{(post_ODCLK)}}$
$f_{\text{PLL CLK}}$	$f_{\text{(GCLK)}}$

Figure 10-4. Operation of the FM-PLL Module

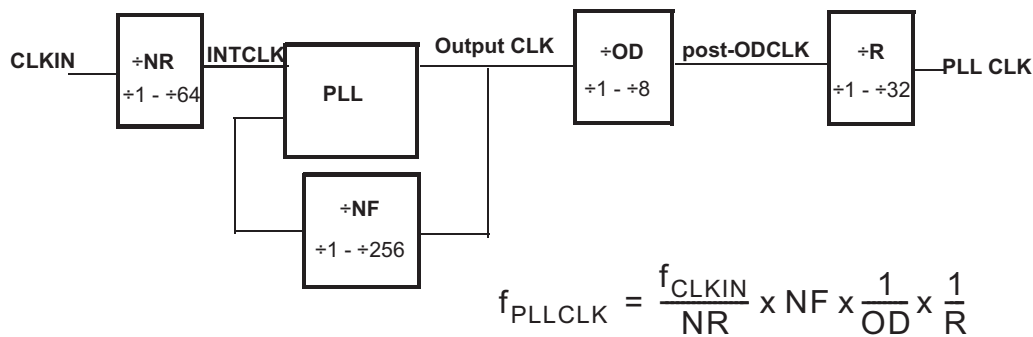


Table 10-2. PLL Value Encoding

	PLL	
NR	$NR = REFCLKDIV[5...0] + 1$	(2)
NF	Non-modulated: $NF = \frac{(PLLMUL[15...0] + 256)}{256}$	(3)
	Modulated: $NF = \frac{(PLLMUL[15...0] + MULMOD[8...0] + 256)}{256}$	(4)
NV	$NV = \frac{(SPR_AMOUNT[8...0] + 1)}{2048}$	(5)
NS	$NS = SPRRATE[8...0] + 1$	(6)
OD	$OD = ODPLL[2...0] + 1$	(7)

NOTE: ODPLL change should occur prior to enabling asynchronous clock domains

Since changing the ODPLL bit-field causes the PLL CLK to be gated, these changes to ODPLL should be completed before configuring a clock domain for an asynchronous clock source. Some clock domains (for example, RTICLK, VCLK2) require a frequency relationship to the VCLK.

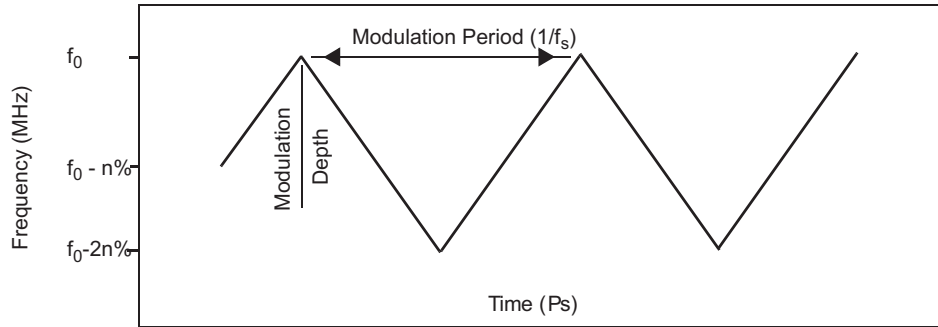
$$f_{VCLK} \geq 3 \times \frac{f_{RTISRC}}{RTIDIV}$$

If the PLL is clocking VCLK and it is stopped for some cycles, then the frequency relationship is temporarily violated.

Many asynchronous domains require frequency relationships between VCLK and the asynchronous domain. Therefore, if the PLL clock is the source for GCLK, HCLK, and VCLK, then the gating produces a short-term change in the PLL clock frequency (and hence also the VCLK frequency). As such, this frequency change could violate the requirements for an asynchronous clock domain.

10.5.1 Modulation

Optionally, the frequency can be modulated, that is, a controlled jitter is introduced onto the baseline frequency of the PLL. This modulation mechanism is not shown in Figure 10-4. When the PLL is used in the modulating mode, the programmable modulation block varies the PLL frequency from the baseline frequency ($f_{\text{baseline}} = (f_{\text{CLKIN}}/NR) \times NF/(OD \times R)$) to $f_{\text{baseline}} \times (1 - 2 \times \text{Depth})$ in a period defined by $1/f_s$; the modulation waveform is triangular and should be enabled after lock.



The modulation is digital and the spreading profile is triangular, down-spread which implies:

- the modulation waveform is composed of a series of frequency steps.
- the modulation frequency and modulation depth are both well controlled due to their digital character.
- the average frequency during modulation is lower than the average frequency prior to enabling modulation. The depth of modulation, however, sets the new average frequency.
- the modulation frequency must be selected slower than the loop bandwidth. From a practical perspective, NS should be near 20.

The modulation fields have a simple geometric meaning:

- the modulation step size is:

$$\frac{NV}{NF} \times f_{\text{OutputCLK}} \quad (8)$$

- the number of steps per modulation period is $2 \times NS$
- the modulation depth is given by:

$$\Delta f = \frac{NS}{2} \times \frac{NV}{NF} \times f_{\text{OutputCLK}}$$

$$Depth [\%] = \frac{NS}{2} \times \frac{NV}{NF} \quad (9)$$

- the modulation frequency is:

$$T_{\text{mod}} = \frac{f_{\text{osc}}}{2 \times NR \times NS} \quad (10)$$

- MULMOD minimizes frequency offset when programmed as:

$$\frac{(SPR_AMOUNT[8..0] + 1)(SPRRATE[8..0] + 1)}{16} \quad (11)$$

NOTE: Modulation should be enabled after Lock

Enable modulation after the lock is completed.

10.5.2 PLL Output Control

The outputs from the PLL are the output clock, slip signals, and VALID.

- **RFSLIP** -- the RFSLIP signal indicates that the Output CLK is running *too fast* relative to INTCLK and sets a RFSLIP status flag in the Global Status Register (GLBSTAT), of the System and Peripheral Control Registers, if the slip signal is active during normal PLL operation; the RFSLIP flag is masked off while the PLL is not active and during the PLL's lock period.
- **FBSLIP** -- the FBSLIP signal indicates that the Output CLK is running *too slow* relative to INTCLK and sets a FBSLIP status flag in the Global Status Register (GLBSTAT), of the System and Peripheral Control Registers, if the slip signal is active during normal PLL operation; the FBSLIP flag is masked off while the PLL is not active and during the PLL's lock period.
- **PLL Slip** -- Logical-OR of the two PLL slip signals. Typically this signal is used to generate a consolidated slip signal to the device (for example, error logic or exception generation). Also used to gate VALID.

NOTE: Clearing Slip Bits

In order to clear any of the slip bits, it is necessary to disable the PLL first.

- **VALID** -- is driven based upon whether the output clock, PLL CLK, is gated or not. However, the VALID signal is dependent upon the PLL Slip signals so that VALID cannot be set if either slip signal is active.
- **PLL Clock** -- The PLL output clock runs at the programmed frequency. When enabled, it takes some time to acquire the programmed frequency (see [Section 10.5.2.1](#)). Similarly, the disable has some timing/constraints (see [Section 10.5.2.2](#)).

10.5.2.1 PLL Enable

After setting the PLL control registers, the clock source is enabled by clearing the appropriate bit in the Clock Source Disable Register (CSDIS) or setting the appropriate bit in the Clock Source Disable Clear Register (CSDISCLR) of the System and Peripheral Control Registers. The bit sends a signal to the PLL that starts the process of enabling the PLL.

1. The PLL checks to make sure that the oscillator is ON. If not, it turns the oscillator ON.
2. The PLL begins a locking process in which the PLL slews from a starting frequency point to the programmed frequency. During this lock period, the PLL slip signals are typically active, and the PLL masks off the signals during this phase. The lock phase takes the following length of time:

Parameter	Value
Lock	$T_{\text{Lock}} = (512 \times T_{\text{OSCIN}}) + (1024 \times \text{NR} \times T_{\text{OSCIN}})$
Enable clocks after lock	$T_{\text{Enable}} = 6 \times T_{\text{OSCIN}}$

3. After the lock phase is complete (when lock counters expire), the PLL releases the slip signals to the system.
4. Then, after the slip signals are released and a delay to enable the clocks, the clock is released to the system and the appropriate CLKSRnV bit for the PLL is set in the Clock Source Valid Status Register (CSVSTAT) of the System and Peripheral Control Registers.

10.5.2.2 PLL Disable

The clock sources (for example, OSC, PLL) are disabled by setting the appropriate bit in the Clock Source Disable Register (CSDIS) or setting the appropriate bit in the Clock Source Disable Set Register (CSDISSET) of the System and Peripheral Control Registers. These bits *allow* the clock to disable but do not force the behavior until the clock is no longer used as the source for a clock domain (for example, GCLK, VCLK, VCLK2, RTICLK).

The PLL receives a signal to disable after the clock is no longer used by any clock domain. Within the PLL, the clock is disabled and the appropriate CLKSRnV bit for the PLL in the Clock Source Valid Status Register (CSVSTAT), of the System and Peripheral Control Registers, becomes inactive. Then the PLL is placed into a low-power state after the following length of time: $T_{\text{Enable}} = 150 \times T_{\text{OSCIN}}$

10.5.2.3 OD-divider Change

The PLL gates the clock if the ODPLL bit-field is changed while the PLL is active. The output clock from the PLL is gated for 3 or 12 OSCIN clock cycles. As the post-ODCLK is gated in the low phase, the output clock to the device -- PLL CLK -- may be gated in a high or low phase though the transition is always glitchless: $T_{\text{ODPLL}} = 3 \times T_{\text{OSCIN}}$

NOTE: ODPLL change should occur prior to enabling asynchronous clock domains

Since changing the ODPLL bit-field causes the PLL CLK to be gated, these changes to ODPLL should be completed before configuring a clock domain for an asynchronous clock source. Some clock domains (for example, RTICLK, VCLK2) require a frequency relationship to the VCLK.

$$f_{\text{VCLK}} \geq 3 \times \frac{f_{\text{RTISRC}}}{\text{RTIDIV}}$$

If the PLL is clocking VCLK and it is stopped for some cycles, then the frequency relationship is temporarily violated.

Many asynchronous domains require frequency relationships between VCLK and the asynchronous domain. Therefore, if the PLL clock is the source for GCLK, HCLK, and VCLK, then the gating produces a short-term change in the PLL clock frequency (and hence also the VCLK frequency). As such, this frequency change could violate the requirements for an asynchronous clock domain.

10.5.2.4 Changing the PLL Operating Point While the PLL is Active

Once the valid bit (CLKSRnV bit in the Clock Source Valid Status Register (CSVSTAT) of the System and Peripheral Control Registers) is set, software may change values to the PLL. If the change of values results in a small percentage change to the VCO frequency ($\Delta f_{\text{OutputCLK}} < 0.1 \times f_{\text{OutputCLK}}$), then these changes can be done on-the-fly. In this mode, the values are updated into the PLL synchronously, and the PLL re-locks to the new value without gating the clocks or the slip bits. If the operating point change is too large, then the slip bits will be set.

Conversely, if the changes to the VCO frequency are large, then the PLL should be disabled prior to changing the values. Typically, any change to the REFCLKDIV field or large changes to the PLLMUL field in the PLL Control Register 1 (PLLCTL1) of the System and Peripheral Control Registers requires a complete disable-and-relock strategy.

10.5.2.5 Summary of PLL Timings

In addition to controlling the lock period and disabling the clock during an ODPLL change, the PLL also generates reset delays. When power-on reset is released (nPORRST 0 --> 1), that release is delayed by 1024 OSCIN cycles so that it is released at the same time that the oscillator valid is asserted. The system reset release is delayed by an additional 8 oscillator clock cycles.

Table 10-3. Summary of PLL Timings

Parameter	Value
nPORRST delay	$T_{nPORRST} = 1024 \times T_{OSCIN}$
nRST delay	$T_{nRST} = 1032 \times T_{OSCIN}$
OSC valid	$T_{OSCVALID} = 1024 \times T_{OSCIN}$
Lock	$T_{Lock} = (512 \times T_{OSCIN}) + (1024 \times NR \times T_{OSCIN})$
Enable clocks after lock	$T_{Enable} = 6 \times T_{OSCIN}$
Disable clocks after lock	$T_{Enable} = 150 \times T_{OSCIN}$
Change ODPLL	$T_{ODPLL} = 3 \times T_{OSCIN}$

10.5.3 Behavior on PLL Fail

The PLL allows flexible response to a PLL failure (slip). Like the oscillator, the PLL clock is configured by default to automatically switch-over to the oscillator in case of a PLL slip. (In this case, the oscillator sources GCM clock source 1 as well as GCM clock source 0. Also, if the oscillator fails, LPO HF is sourced to both GCM clock sources 0 and 1.)

The PLL slip outputs indicate that the PLL is running either too fast or too slow. These error output toggle when the PLL is locking and when the PLL is disabling. The PLL blocks these slip outputs during these times, leaving them active only while the PLL is active.

A slip after the PLL has locked and while it is active is an indication of a PLL failure. The PLL provides slip-filtering which enhances the flexibility of the PLL's response to failure. The slip-filtering circuit samples the slip based on HF LPO. The filter defines the number of consecutive HF LPO cycles for which the slip signal must be active before the slip is recognized. This slip is latched in the RFSLIP and FBSLIP status flags in the Global Status Register (GLBSTAT) of the System and Peripheral Control Registers.

The PLL may enable/disable the automatic switch over as well as the error signaling; if the error signaling is enabled, a PLL slip may be configured to generate a reset. The automatic switch-over and suppression of the error signals are controlled by the bypass on slip bit field -- BPOS[1:0] (PLLCTL1.(30:29)). When BPOS[1:0] is disabled (BPOS[1:0] = 10b):

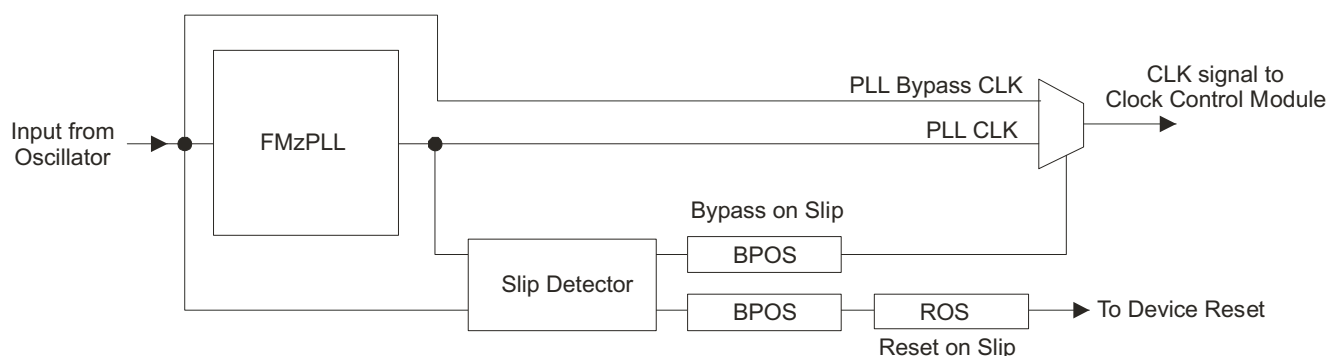
- automatic response to the PLL slip is prevented
- ESM/exception is NOT generated
- reset on slip is not generated regardless of the state of the ROS bit
- status bits are set on a PLL slip independent of BPOS[1:0]

When BPOS[1:0] is enabled (BPOS[1:0] = 00b OR 01b OR 11b):

- PLL slip causes the clock source into GCM clock source 1 to shift from the PLL to the oscillator
- ESM/exception is generated
- reset on slip is generated if ROS is set

The effect of BPOS[1:0] on the system is shown in [Figure 10-5](#).

Figure 10-5. PLL Slip Detection and Reset/Bypass Block Diagram



10.5.4 Recovery from a PLL Failure

If PLL1 fails, the PLL's slip causes the valid flag to be locked and causes the clock source into GCM clock source 1 to shift from the PLL to the oscillator. The RFSLIP or FBSLIP status flags in the Global Status Register (GLBSTAT) of the System and Peripheral Control Registers are also set. PLL1 may be re-enabled (though if the failure was caused by a hard-fault, the re-enable will fail) through the following procedure:

1. Switch all clock domains from PLL1 to the oscillator (for example, GHVSRG uses oscillator, VCLKAn uses oscillator or VCLK, and so on).
2. Disable PLL1 with CSDISSET. This action disables the PLL and causes the slip signal to no longer be driven. Valid is not released until the slip is cleared.
3. Clear the RFSLIP or FBSLIP status flags in the Global Status Register (GLBSTAT) of the System and Peripheral Control Registers by writing a 1 to the bit. After this step, the valid flag is unlocked and cleared if it was previously set.
4. Re-enable PLL1 with CSDISCLR.
5. Switch the clock domains back to PLL1.

If PLL2 fails, the PLL's slip causes the valid flag to be locked. There is no autonomous change of clock source for PLL2. Neither the RFSLIP or FBSLIP status flags in the Global Status Register (GLBSTAT) of the System and Peripheral Control Registers are set. PLL2 may be re-enabled in a similar procedure to re-enabling PLL1 (though if the failure was caused by a hard-fault, the re-enable will fail):

1. Switch all clock domains from PLL2 to the oscillator (for example, GHVSRG uses oscillator, VCLKAn uses oscillator or VCLK, and so on).
2. Disable PLL2 with CSDISSET. This action disables the PLL and causes the slip signal to no longer be driven. Valid is not released until the slip is cleared.
3. Reset PLL2 Valid by writing a 1 to both RFSLIP and FBSLIP status flags in the Global Status Register (GLBSTAT) of the System and Peripheral Control Registers (even though they are not set by the slip). After this step, the valid flag is unlocked and cleared if it was previously set.
4. Re-enable PLL2 with CSDISCLR.
5. Switch the clock domains back to PLL2.

10.5.5 PLL Modulation Depth Measurement

The PLL contains a circuit for estimating the depth of the modulation. The circuit counts clock edges over a fixed window of the modulation waveform (SSW_CAPTURE_COUNT in SSWPLL2) and clock edges over the entire waveform (SSW_CLKOUT_COUNT in SSWPLL3). The capture ends after a pre-determined number of clock edges in SSW_CLKOUT_COUNTER as set in TAP_COUNTER_DIS. There are $2 \times NR$ windows per modulation waveform. The procedure for estimating the modulation depth is:

1. While GCLK is sourced by the oscillator and the PLL is enabled with modulation, configure SSWPLL1 as follows:
 - a. CAPTURE_WINDOW_INDEX is set equal to NR.
 - b. COUNTER_RESET is set.
 - c. TAP_COUNTER_DIS is set to disable the measurement after SSW_CLKOUT_COUNT captures this number of clocks. The measurement is disabled after the set tap is set AND the modulation cycle ends.
 - d. Ensure that EXT_COUNTER_EN is cleared.
2. Ensure that both SSW_CAPTURE_COUNT and SSW_CLKOUT_COUNT are cleared (by the COUNTER_RESET).
3. Set COUNTER_EN and clear COUNTER_RESET. This step releases the reset and enables the counter to begin counting.
4. After a wait loop, poll for COUNTER_READ_READY to set. After the bit is set, read SSW_CAPTURE_COUNT and SSW_CLKOUT_COUNT.
5. Compute the modulation depth as:

$$Depth = abs\left(1 - \frac{2 \times NR \times SSW_CAPTURE_COUNT}{SSW_CLKOUT_COUNT}\right) \quad (12)$$

10.5.6 PLL Frequency Measurement Circuit

The same circuit that is used to measure modulation depth is also available to measure the average frequency of the PLL. In this mode, the PLL output (before the R-divider) is captured in SSW_CLKOUT_COUNT while the oscillator is captured in SSW_CAPTURE_COUNT. The procedure for using the PLL frequency measurement circuit is:

1. While the PLL is enabled, set EXT_COUNTER_EN.
2. Set COUNTER_EN. This bit clears both SSW_CAPTURE_COUNT and SSW_CLKOUT_COUNT and then immediately enables for counting.
3. Wait for some software delay loop.
4. Clear COUNTER_EN. Wait for COUNTER_READ_READY to set. Read both SSW_CAPTURE_COUNT and SSW_CLKOUT_COUNT and compute the ratio of PLL multiplication as:

$$\frac{NF}{NR \times OD} = \frac{SSW_CLKOUT_COUNT}{SSW_CAPTURE_COUNT} \quad (13)$$

5. Note that CAPTURE_WINDOW_INDEX, COUNTER_RESET, TAP_COUNTER_DIS are not used in this procedure.

10.5.7 PLL2

PLL2 drives GCM clock source 6.

The PLL is identical to PLL1, except modulation is disabled on this instance of the PLL. Also, the PLL typically does not clock the system, there is no automatic switch over feature. Any PLL error can be handled by the CPU.

PLL2 is programmed through PLLCTL3.

10.6 PLL Control Registers

The clock module has two registers (PLLCTL1 and PLLCTL2) located within the System and Peripheral Control Registers, plus it has four bits located in other System and Peripheral Control Registers.

The FM-PLL is off at power-on. The clock source is enabled by clearing the appropriate bit in the Clock Source Disable Register (CSDIS) or setting the appropriate bit in the Clock Source Disable Clear Register (CSDISCLR) of the System and Peripheral Control Registers. [CSDISCLR and Clock Source Disable Set Register (CSDISSET) also enable/disable the PLL and oscillator (and other clock sources).]

The LPOCLKDET module generates the OSCFAIL flag in the Global Status Register (GLBSTAT), of the System and Peripheral Control Registers, if a problem with the reference oscillator is detected. The slip signals are also registered in the RFSLIP and FBSLIP status flags in the Global Status Register (GLBSTAT), of the System and Peripheral Control Registers, in order to indicate the source of a clock failure.

The appropriate CLKSRnV bit for the PLL is set in the Clock Source Valid Status Register (CSVSTAT) of the System and Peripheral Control Registers.

The following sections describe the PLL registers used in the system module. These registers support 8, 16, and 32-bit write accesses. The reset values for these registers are configured so that an input frequency in the range from 5MHz to 20MHz generates a valid clock.

Table 10-4. PLL Module Registers

Offset	Acronym	Register Description	Section
FFFF FF30h	CSDIS	Clock Source Disable Register	Section 2.5.1.10
FFFF FF34h	CSDISSET	Clock Source Disable Set Register	Section 2.5.1.11
FFFF FF38h	CSDISCLR	Clock Source Disable Clear Register	Section 2.5.1.12
FFFF FF54h	CSVSTAT	Clock Source Valid Status Register	Section 2.5.1.19
FFFF FF70h	PLLCTL1	PLL Control 1 Register	Section 2.5.1.26
FFFF FF74h	PLLCTL2	PLL Control 2 Register	Section 2.5.1.27
FFFF E100h	PLLCTL3	PLL Control 3 Register	Section 2.5.2.1
FFFF FFA0h	GPREG1	General Purpose Register	Section 2.5.1.35
FFFF FFECh	GLBSTAT	Global Status Register	Section 2.5.1.51
FFFF E170h	CLKSLIP	PLL Clock Slip Control Register	Section 2.5.2.5
FFFF FF24h	SSWPLL1	PLL Modulation Depth Measurement Control Register	Section 10.6.1
FFFF FF28h	SSWPLL2	SSW PLL BIST Control Register 2	Section 10.6.2
FFFF FF2Ch	SSWPLL3	SSW PLL BIST Control Register 3	Section 10.6.3

Table 10-5. LPOCLKDET Module Registers

Offset	Acronym	Register Description	Section
FFFF FF88h	LPOMONCTL	LPO/Clock Monitor Control Register	Section 2.5.1.31
FFFF FF8Ch	CLKTEST	Clock Test Register	Section 2.5.1.32

10.6.1 PLL Modulation Depth Measurement Control Register (SSWPLL1)

Figure 10-6 illustrates this register and Table 10-6 provides the bit descriptions. This register applies to PLL1, but does not apply to PLL2.

Figure 10-6. SSW PLL BIST Control Register 1 (SSWPLL1) [offset = FF24h]

31	Reserved																16
R-0																	
15	CAPTURE_WINDOW_INDEX																8
R/W-0																	
7	6	5	4	3	1						0						
Reserved	COUNTER_READ_READY	COUNTER_RESET	COUNTER_EN	TAP_COUNTER_DIS						EXT_COUNTER_EN							
R-0	R-0	R/W-1	R/W-0	R/W-0						R/W-0							

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 10-6. SSW PLL BIST Control Register 1 (SSWPLL1) Field Descriptions

Bit	Field	Value	Description
31-16	Reserved	0	Read returns 0. Writes have no effect.
15-8	CAPTURE_WINDOW_INDEX	0-FFh	The capture counter present in the PLL wrapper will count the PLL clock edges when the current modulation phase capture window value is equal to these bits. Should be set equal to NR.
7	Reserved	0	Read returns 0. Writes have no effect.
6	COUNTER_READ_READY	0 1	Counter read ready. Indicates that SSW_CAPTURE_COUNT (SSWPLL2) and SSW_CLKOUT_COUNT (SSWPLL3) can be read. 0 Counter registers in SSWPLL2 and SSWPLL3 are not ready to read. 1 Counter registers in SSWPLL2 and SSWPLL3 are ready to read.
5	COUNTER_RESET	0 1	Counter reset. If EXT_COUNTER_EN = 0, COUNTER_RESET resets SSW_CAPTURE_COUNT (SSWPLL2) and SSW_CLKOUT_COUNT (SSWPLL3). If EXT_COUNTER_EN = 1, this bit is ignored. 0 No impact to counters 1 If the EXT_COUNTER_EN bit is 0, then counters SSW_CAPTURE_COUNT and SSW_CLKOUT_COUNT will be held in the reset state. If EXT_COUNTER_EN bit is 1, then this bit will be ignored by the PLL wrapper.
4	COUNTER_EN	0 1	Counter enable. If EXT_COUNTER_EN = 0, COUNTER_EN initializes the modulation depth measurement. (In this mode, the disable is set to occur automatically.) If EXT_COUNTER_EN = 1, the counters are enabled/disabled with COUNTER_EN. 0 If EXT_COUNTER_EN = 0, COUNTER_EN = 0 indicates that the counters are inactive. If EXT_COUNTER_EN = 1, COUNTER_EN = 0 disables the counters. 1 If EXT_COUNTER_EN = 0, COUNTER_EN = 1 indicates that the counters are still active. If EXT_COUNTER_EN = 1, COUNTER_EN = 1 enables the counters.

Table 10-6. SSW PLL BIST Control Register 1 (SSWPLL1) Field Descriptions (continued)

Bit	Field	Value	Description
3-1	TAP_COUNTER_DIS	0 Bit 16 of CLKOUT counter is selected. When this bit is set and the modulation period finishes, the counters are disabled and READ_READY_FLAG is set. 1h Bit 18 of CLKOUT counter is selected. 2h Bit 20 of CLKOUT counter is selected. 3h Bit 22 of CLKOUT counter is selected. 4h Bit 24 of CLKOUT counter is selected. 5h Bit 26 of CLKOUT counter is selected. 6h Bit 28 of CLKOUT counter is selected. 7h Bit 30 of CLKOUT counter is selected.	The value in this register is used to program a particular bit in CLKOUT counter. When that particular bit in CLKOUT counter becomes 1, then both the CLKOUT counter and the CAPTURE counter will stop counting when EXT_COUNTER_EN = 0. When EXT_COUNTER_EN = 1, this bit field is not used.
0	EXT_COUNTER_EN	0 Modulation Depth Measurement mode 1 Frequency Measurement mode	

10.6.2 SSW PLL BIST Control Register 2 (SSWPLL2)

This is an observation register used to log counter value for the capture counter inside the PLL wrapper. This register applies to PLL1, but does not apply to PLL2. The SSWPLL2 register is shown in [Figure 10-7](#) and described in [Table 10-7](#).

Figure 10-7. SSW PLL BIST Control Register 2 (SSWPLL2) [offset = FF28h]

31	SSW_CAPTURE_COUNT	16
	R-0	
15	SSW_CAPTURE_COUNT	0
	R-0	

LEGEND: R = Read only; -n = value after reset

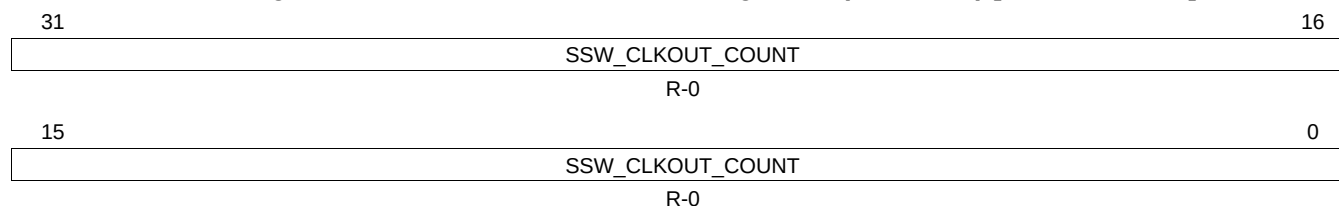
Table 10-7. SSW PLL BIST Control Register 2 (SSWPLL2) Field Descriptions

Bit	Field	Value	Description
31-0	SSW_CAPTURE_COUNT	0-FFFF FFFFh	Capture count. This register returns the value of the capture count. When EXT_COUNTER_EN = 0, this counter increments within a fixed modulation window. When EXT_COUNTER_EN = 1, this counter increments based upon the oscillator.

10.6.3 SSW PLL BIST Control Register 3 (SSWPLL3)

This is observation register used to log counter value for CLKOUT counter inside PLL wrapper. This register applies to PLL1, but does not apply to PLL2. The SSWPLL3 register is shown in [Figure 10-8](#) and described in [Table 10-8](#).

Figure 10-8. SSW PLL BIST Control Register 3 (SSWPLL3) [offset = FF2Ch]



LEGEND: R = Read only; -n = value after reset

Table 10-8. SSW PLL BIST Control Register 3 (SSWPLL3) Field Descriptions

Bit	Field	Value	Description
31-0	SSW_CAPTURE_COUNT	0-FFFF FFFFh	Value of CLKout count register. This counter increments based upon the PLL output (prior to the R-divider).

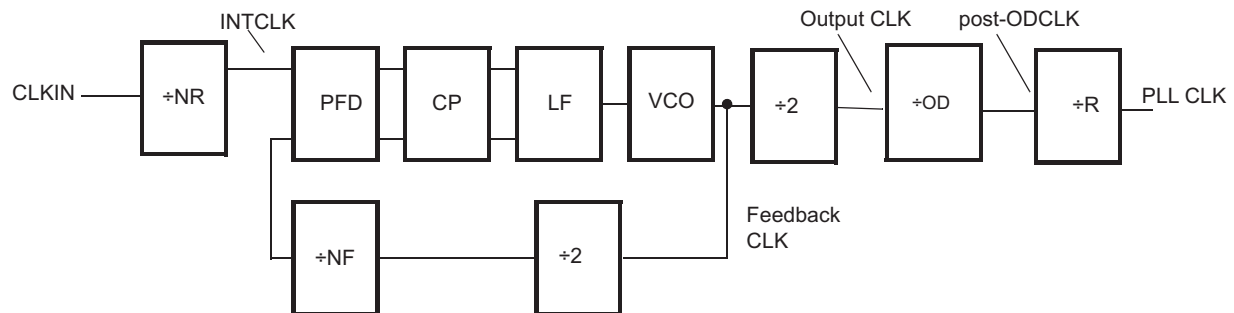
10.7 Phase-Locked Loop Theory of Operation

The PLL block consists of six logical sub-blocks:

- Phase-Frequency Detector (PFD)
- Charge Pump (CP)
- Loop Filter (LF)
- Voltage-Controlled Oscillator (VCO)
- Frequency Modulation
- Slip Detector

Figure 10-9 illustrates the sub-blocks in a basic PLL circuit. The VCO adjusts its frequency until the two signals into the PFD have the same phase and frequency. The feedback path (from VCO to PFD) divides the frequency of the feedback signal by $2 \times NF$; this feedback divider requires the VCO to generate a frequency $2 \times NF$ times greater than the internal frequency ($OSCIN/NR$). In the forward path (from VCO to PLL CLK), the $/2$ block creates a clean duty cycle.

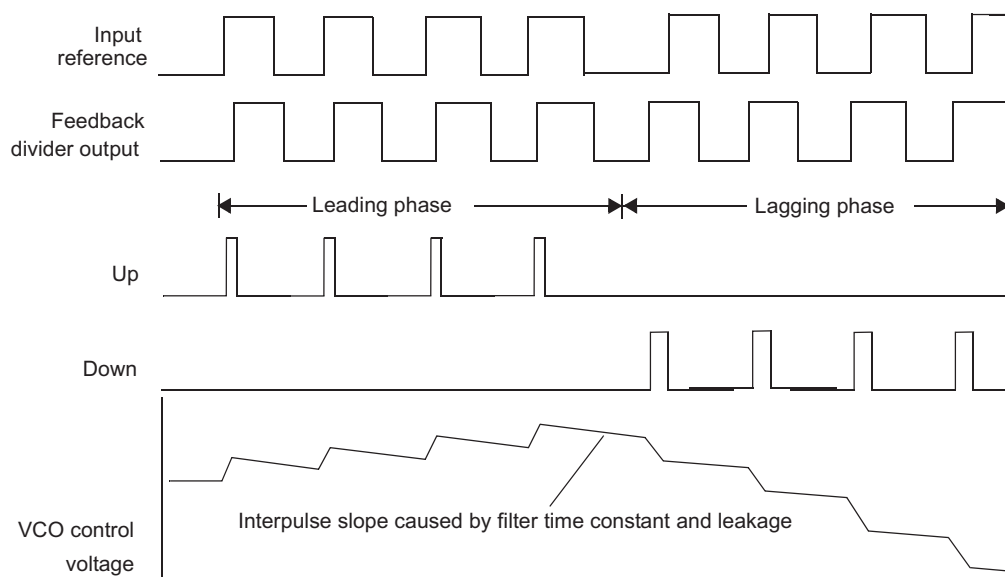
Figure 10-9. Basic PLL Circuit



10.7.1 Phase-Frequency Detector

The phase-frequency detector (PFD) compares the input reference phase/frequency to the phase/frequency of the feedback divider and generates two signals: an *up* pulse and a *down* pulse that drive a charge pump. The resulting charge, when integrated by the circuit at the LF pin, provides a VCO control voltage, as shown in Figure 10-10.

Figure 10-10. PFD Timing



The width of the up pulse and the down pulse depends on the difference in phase between the two inputs. For example, when the reference input leads the feedback input by 10 ns, then an up pulse of approximately 10 ns is generated (see [Figure 10-10](#)). On the other hand, when the reference input lags the feedback input by 10 ns, then a down pulse of approximately 10 ns is generated. When the two inputs are exactly in phase, the up pulse and down pulse become essentially zero-width. These pulses are fed to the charge pump block, which meters charge into the low-pass loop filter.

The advantage of a phase-frequency detector over a phase-only detector is that it cannot lock to a harmonic or subharmonic of the reference. This important property also ensures that the output frequency of the VCO is always exactly $2 \times NF$ times the reference frequency.

The reference feedback frequency is based upon the VCO frequency and the feedback divider. Fractional multiplication is achieved by changing the feedback divider real-time in order to create the fractional multiplication. As an example, if a multiplier of 100.5 is selected, the feedback divider divides by 100 and 101 in equal proportions; in this case, the PLLMUL bit field would be programmed as 99.5 (0x6380). This fractional multiplication is useful when trying to achieve final frequencies that are non-integer to the input frequency (for example, a final frequency that is a prime number). The fractional portion of the divider should be small compared to the multiplier and so it is recommended that the fractional portion relate to parts in 16, implying that the last 4 bits should always be 0.

10.7.2 Charge Pump and Loop Filter

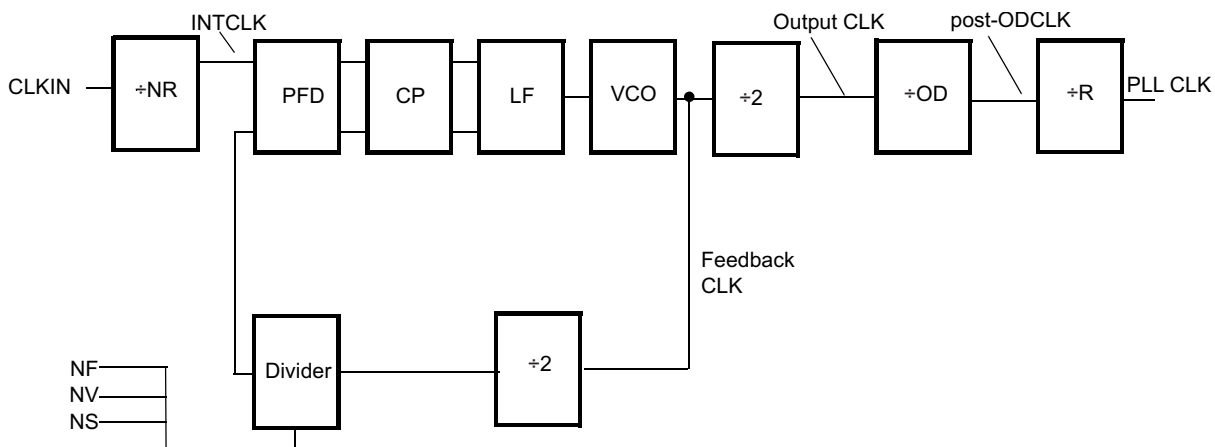
The charge pump (CP) add or remove charge from the loop filter based on the pulses coming from the phase-frequency detector (PFD).

Two components of the filter output signal are summed together: an integral component and a proportional component. The integral component maintains a DC level going to the VCO to set its frequency, and the proportional component makes the VCO track changes in phase to minimize jitter. The capacitors and resistors required for the filter are integrated in silicon.

10.7.3 Voltage-Controlled Oscillator

The output frequency of the VCO is proportional to its input control voltage, which is generated by the charge pump via the integrated loop filter. If the VCO oscillates too slowly, the feedback phase begins to lag the reference phase at the PFD, which increases the control voltage at the VCO. Conversely, if the VCO oscillates too fast, the feedback phase begins to lead the reference phase at the PFD, which decreases the control voltage at the VCO. These two actions keep the VCO running at the correct frequency multiple of the reference.

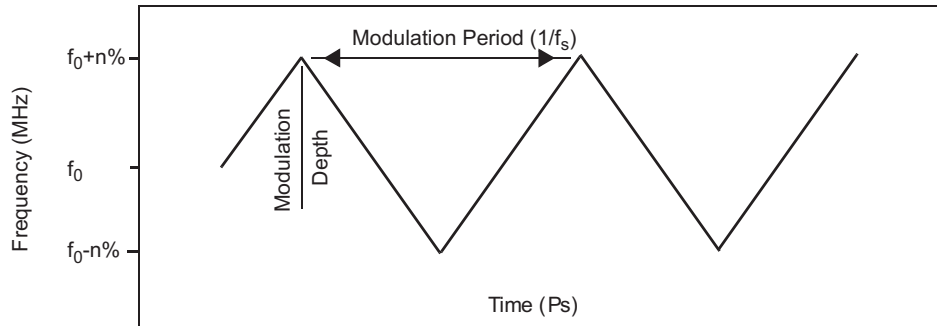
Figure 10-11. PLL Modulation Block Diagram



10.7.4 Frequency Modulation

The output clock of the PLL changes frequency in a controlled way, centered around the unmodulated output frequency. The modulation block directly modulates the VCO frequency at the loop filter, and creates the triangular frequency modulation (see Figure 10-12).

Figure 10-12. Frequency vs. Time



10.8 Programming Example

This section provides an example of how to program the PLL. For non-modulation settings, the PLLCTL1 and PLLCTL2 settings from 130nm process devices can be used without modification.

Suppose that, using a 20MHz crystal, the application requires:

- 180MHz GCLK (and HCLK) frequency
- 100 kHz spreading frequency
- 0.5% spreading depth

1. Choose an NR and NS such that:

$$\bullet \quad \frac{f_{CLKIN}}{NR \times f_s} \geq 40 \quad (14)$$

$$\bullet \quad f_s \equiv \frac{f_{CLKIN}}{2 \times NR \times NS} \quad (15)$$

$$\bullet \quad 2 \times NS = \frac{f_{CLKIN}}{NR \times f_s} \geq 40 \quad (16)$$

- (NR,NS) = {(5,20), (4,25), (2,50), (1,100)}

- Either NR = 5 and NS = 20 or NR = 4 and NS = 25 are reasonable. Another choice (NR = 3 and NS = 33) is possible, if the modulation frequency can vary from 100KHz.

2. Choose Output CLK frequency as integer divider of output frequency near to 330MHz. Output CLK frequency shall not exceed 550MHz or fall below 150MHz.

The integer values for 180MHz are 360MHz or 540MHz. 360MHz is close to the target frequency of 330MHz and we use this frequency.

3. In this case, either of the following equations are suitable choices for getting to 360MHz. Choose NR = 5, NS = 20 and set NF = 90 or choose NR = 4, NS = 25 and set NF = 72.

$$\frac{f_{CLKIN}}{NR} = \frac{20[\text{MHz}]}{4} = 5[\text{MHz}] \quad \text{or} \quad \frac{f_{CLKIN}}{NR} = \frac{20[\text{MHz}]}{5} = 4[\text{MHz}] \quad (17)$$

4. Select the output divider OD so that the post-ODCLK frequency does not exceed the maximum frequency of output divider R (device specific frequency). In this case, choose OD = 2 and R = 1.
5. Compute the divider value NV:

$$Depth = \frac{0.5}{100} = \frac{NV}{NF} \times \frac{NS}{2} = \frac{NV}{90} \times \frac{20}{2} \quad (18)$$

$$NV = 0.045$$

6. If it is important to maintain the same average frequency in modulation as in non-modulation, either NF should be modified OR program the MULMOD bit field. The modulation fields create a multiplier offset equal to:

$$\Delta NF = \frac{NV \times NS}{2} \quad (19)$$

If using MULMOD[8:0], then:

$$\Delta NF = \frac{MULMOD[8...0]}{256} = \frac{NV \times NS}{2} = \frac{0.045 \times 20}{2} \quad (20)$$

$$MULMOD[8...0] = \frac{0.045 \times 20}{2} \times 256 = 115.2 \quad (21)$$

MULMOD will be set to 115.

7. Convert the PLL parameters into bit field values:
 - NR = 5, implies that REFCLKDIV[5:0] = 4
 - NS = 20, implies that SPRATE[8:0] = 19 = 0x13
 - NF = 90, implies that PLLMUL[15:0] = 0x5900
 - OD = 2, implies that ODPLL[2:0] = 1
 - R = 1, implies that PLLDIV[4:0] = 0
 - NV = 0.045, implies that SPR_AMOUNT[8:0] = 91 = 0x5B
 - MULMOD[8:0] = 115 = 0x73
8. Setting only these fields (that is, not BPOS, ROF, or ROS) yields:

PLLCTL1 = 0x00045900

PLLCTL2 = 0x04C7325B

When FM ENA is turned on, PLLCTL2 = 0x84C7325B

The Output CLK is centered in the range from 150MHz to 550MHz at 360MHz.

NF = 90 falls within the multiplier range from 1 to 256.

OD is selected so that post-ODCLK meets the device specification.